

VENKATESAN MURALI

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Summary

Senior Software Engineer with 10+ years of experience designing and building scalable cloud solutions using AWS, Kubernetes, and Kafka. Specialized in architecting AI-driven systems leveraging OpenAI, Bedrock, and modern event-driven architectures. Passionate about driving engineering excellence and acting as a force multiplier by empowering teams to deliver robust, enterprise-grade solutions at scale.

Education

Master of Science, Computer science – GPA 3.93 / 4

Arizona State University, Tempe, United States

Bachelor of Engineering, Computer Science and Engineering - GPA 8.01 / 10

Anna University, Chennai, India

Technical Skills

Programming Languages and Frameworks	: C#, .Net, TypeScript, JavaScript, Node.js
Frontend Frameworks and Visualization	: Angular, React, D3.js
Event Streaming	: Apache Kafka managed by Confluent Cloud
Cloud and DevOps	: AWS (EKS, ECS, S3, Lambda, IAM, Cloudfront, SQS, SNS, Route 53), Terraform, Google Cloud (Dialogflow, , Google Cloud Storage, Google Cloud Functions)
AI & LLM Integration	: OpenAI APIs, Gemini Models, AWS Bedrock
Conversational AI	: Google DialogFlow, Amazon Lex
Databases & Data Stores	: Amazon RDS (PostgreSQL, SQL Server), DynamoDB, Redis, Valkey
Monitoring & Observability	: Dynatrace, Splunk, AWS Cloudwatch, Pagerduty
Testing & Automation Framework	: Cypress, Playwright, xUnit, WireMock, Moq, Jasmine
AI-powered Development Tools	: Github Copilot, Claude Code, Kiro
Design and Architecture Tools	: Lucidchart, Signavio (architecture and system diagrams), Figma (UI mockups)
CI/CD & Automation	: CircleCI, GitHub Actions

Professional Experience

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| Senior Software Engineer | Rocket Mortgage | Sep 2023 – Present |
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- Designed and led the development of a Voice AI orchestration platform that automated 50% of outbound lead calls, dynamically generating conversation flows based on lead type (e.g., refinance or purchase). Collaborated on system architecture and integrated Twilio for multi-leg call routing, enabling seamless banker handoffs, reducing manual effort by 50%, and doubling lead engagement scalability. **Technologies used:** AWS EKS, .Net, PostgreSQL (analytics), Valkey (operational data), Twilio, various LLM models for voice AI.
 - **Automated multilingual localization** for the Rocket Mortgage Servicing Website by building an EKS pipeline triggered via CircleCI that translates English content (en.json) to Spanish (es.json) using a **champion-challenger model** to select the best LLM (OpenAI, Gemini, Anthropic) based on BLEU scores, enabling scalable translations and reducing manual effort. **Technologies used:** EKS, .Net, Angular localization using i18n, CircleCI triggers, Auth0, LLM Integrations with Gemini, OpenAI and Anthropic.
 - Architected and enhanced the Rocket Mortgage Servicing chat application by **integrating generative AI with Bedrock Agents** for orchestration, enabling intelligent transfers to experts, bankers, and Kendra-powered knowledge bases. This improvement **reduced containment rates by 30%** and leveraged **Nest.js** backend, **Angular Micro Frontend UI**, **AWS Lambda**, and **Kendra** for seamless AI-driven support.
 - **Mentored and guided** multiple engineers across teams, fostering technical growth and best practices, and played a key role in helping several engineers achieve promotions.
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| Software Engineer | Rocket Mortgage | June 2019 – Sep 2023 |
|--------------------------|------------------------|----------------------|
- **Optimized** the [Rocket Mortgage Servicing](#) landing page by implementing lightweight Angular web components and refactoring data flow using async pipes and selective API calls, resulting in a 40% improvement in page load time and enhanced user experience. **Technologies used:** Angular, MicroServices, Micro FrontEnds, Async Programming, Node.js, Playwright.

- **Implemented** comprehensive observability for Rocket Mortgage Servicing by integrating Dynatrace metrics across backend and frontend, adding anomaly detection, PagerDuty alerting, real user monitoring, and synthetic monitoring. This proactive approach **improved client experience** through faster issue detection and resolution.
- **Pioneered** the use of React microfrontend within an Angular host using Module Federation by building a Purchase Calculator that enabled clients to estimate monthly mortgage payments with tax and insurance options. This innovation **enhanced user experience** and drove significant traffic growth on the Rocket Mortgage Servicing site. **Technologies used:** Module Federation, React, Angular, Cypress, Jasmine.

Web Developer Co-Op

Token IQ, Inc

Feb 2019 – May 2019

- Developing a platform for fundraising (IPOs and ICOs) where Issuers and Investors can interact and exchange funds through coins or currencies. **Technologies used:** Angular, JavaScript, Stellar Network for Blockchain, Node.js.

Student Intern (Full Stack Developer)

Esri

May 2018 – Aug 2018

- Collect, store and analyze statistics of ArcGIS Enterprise Data Store system like CPU, memory, disk IO and network usage, analyze performance in data store's underlying databases like PostgreSQL and Elasticsearch. Built a rest endpoint to expose the statistics result and displayed the results in a visualization tool in the Web UI using charts. **Technologies used:** Java, Highcharts.js, Jersey, SQLite, PostgreSQL.

Web Applications Developer

EdPlus at Arizona State University

Jan 2018 – Jan 2019

- Developed single page applications using React and integrated into edX studio to provide enriching course experience to students. **Technologies used:** React.js.
- Connected Marketing Cloud with Salesforce instances to sync student data. **Technologies used:** Java, PostgreSQL, REST API

Senior Software Engineer

Lister Technologies Pvt Ltd

July 2015 – July 2017

- Implemented the [support portal](#) for Rosetta Stone Inc. which works in 13 different languages that answered most product related questions without any assistance from the support center. Consequently, this saved the company a lot of money by eliminating labor and call center costs. **Technologies used:** Force.com Sites, VisualForce pages, Apex, JavaScript, Bootstrap.
- Developed a self-checkout page with credit card authorization for an online health and legal service organization to automate their inventory management and purchases. **Technologies used:** JavaScript, REST API.

Software Engineer

Lister Technologies Pvt Ltd

July 2013 – June 2015

- Implemented real time scalable integrations from a native inventory management system to Salesforce Sales Cloud for capturing product related data in Salesforce for Forecasts. **Technologies used:** Java, Salesforce REST API.

Academic Projects

Content and Collaborative Filtering Based Recommendation Engine for Stack Overflow

Aug 2018 – Dec 2018

- Developed a web application to log user activity from stackoverflow.com, analyze the data collected, visualize the data and recommend crawled content based on user search history. **Technologies used:** Spring Boot, Java, React JS, MongoDB.

Geospatial Hotspot Analysis of New York City Taxi Usage

Aug 2018 – Dec 2018

- Built a system to analyze the taxi usage pattern of New York city using Getis-Ord statistics. Implemented spatial operations and hotspot detection using a distributed spark cluster. **Technologies used:** Scala, Apache Spark.

Human Identification Using Brain Waves

Jan 2018 – May 2018

- Designed an application to detect humans using their brain waves. Developed an android application that executes the machine learning algorithm in the phone and on the cloud server. **Technologies used:** Java, Python, Android, AWS.